

Sri Lanka Institute of Information Technology

B.Sc. (Hons) Information Technology



Specialization in Information Technology

IT3020 - Database Systems

Laboratory Worksheet 05

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```
1  SET SERVEROUTPUT ON;
2
3  DECLARE
4  BEGIN
5      DBMS_OUTPUT.PUT_LINE('This is my first PL/SQL Program');
6  END;
7  /
```

Output:

Query result Script output **DBMS output** Explain Plan SQL history

 

This is my first PL/SQL Program  

Exercise 01

01. Write a PL/SQL to get the current price of a stock belongs to company 'IBM'. Store the company name ('IBM') in a variable.

```
9  --Exercise 01
10 --Get the current price of stock 'IBM'. Store company name in a variable.
11
12 DECLARE
13     v_company  VARCHAR2(20) := 'IBM';
14     v_price    NUMBER(8,2);
15 BEGIN
16     SELECT current_price
17     INTO v_price
18     FROM stocks_obj_tab
19     WHERE company = v_company;
20
21     DBMS_OUTPUT.PUT_LINE('Company : ' || v_company);
22     DBMS_OUTPUT.PUT_LINE('Current Price : ' || v_price);
23 END;
24 /
~
```

Output:

Query result	Script output	DBMS output	Explain Plan	SQL history
 				
   				
Company : IBM Current Price : 70				

Exercise 02

02. Write a PL/SQL to display a message for a given stock of 'IBM' company according to the following criteria.

- Current price < 45 - 'Current price is very low!'
 - 45<=Current price < 55 - 'Current price is low!'
 - 55<=Current price < 65 - 'Current price is medium!'
 - 65<=Current price < 75 - 'Current price is medium high!'
 - 75<=Current price - 'Current price is high!'
- Store the company name in a variable.

```

26  --Exercise 02
27  --Display message according to IBM price range
28
29  DECLARE
30      v_company  VARCHAR2(20) := 'IBM';
31      v_price     NUMBER(8,2);
32  BEGIN
33      SELECT current_price
34      INTO v_price
35      FROM stocks_obj_tab
36      WHERE company = v_company;
37
38      IF v_price < 45 THEN
39          DBMS_OUTPUT.PUT_LINE('Current price is very low !');
40      ELSIF v_price >= 45 AND v_price < 55 THEN
41          DBMS_OUTPUT.PUT_LINE('Current price is low !');
42      ELSIF v_price >= 55 AND v_price < 65 THEN
43          DBMS_OUTPUT.PUT_LINE('Current price is medium !');
44      ELSIF v_price >= 65 AND v_price < 75 THEN
45          DBMS_OUTPUT.PUT_LINE('Current price is medium high !');
46      ELSE
47          DBMS_OUTPUT.PUT_LINE('Current price is high !');
48      END IF;
49  END;
50  /

```

Output:

Query result	Script output	DBMS output	Explain Plan	SQL history
 				
Current price is medium high !				

Exercise 03

03. Write three PL/SQLs to draw the following shape. Use all three types of loops in each PL/SQL


 PL_SQL_Lab05.sql
 




 Aa
 

```

52  --Exercise 03
53  --Draw shape using three types of loops
54
55  --1.Simple Loop
56  BEGIN
57      FOR i IN REVERSE 1..9 LOOP
58          DECLARE
59              j NUMBER := i;
60          BEGIN
61              LOOP
62                  DBMS_OUTPUT.PUT(i || ' ');
63                  j := j + 1;
64                  EXIT WHEN j > 9;
65              END LOOP;
66              DBMS_OUTPUT.NEW_LINE;
67          END;
68      END LOOP;
69  END;
70  /

```

```

89  --3.For Loop
90  BEGIN
91      FOR i IN REVERSE 1..9 LOOP
92          FOR j IN 1..9 LOOP
93              DBMS_OUTPUT.PUT(i || ' ');
94          END LOOP;
95          DBMS_OUTPUT.NEW_LINE;
96      END LOOP;
97  END;
98  /
99

```

Output:

Query result
Script output
DBMS output
Explain Plan
SQL history




```

9
88
777
6666
55555
444444
3333333
22
11
72  --2.While Loop
73  DECLARE
74      i NUMBER := 9;
75      j NUMBER;
76  BEGIN
77      WHILE i >= 1 LOOP
78          j := 9;
79          WHILE j >= i LOOP
80              DBMS_OUTPUT.PUT(i || ' ');
81              j := j - 1;
82          END LOOP;
83          DBMS_OUTPUT.NEW_LINE;
84          i := i - 1;
85      END LOOP;
86  END;
87  /

```





Exercise 04

04. The companies have decided to offer a bonus to the loyal clients who have purchased stocks of their company. A client who has purchased stocks before 1st January 2002 gets an additional 50 stocks as bonus. Those who have purchased stocks before 1st January 2001 get 100 stocks and for the purchases done before 1st January 2000 will be given 150 stocks as bonus.

Write a PL/SQL block to update the purchase table. Increase the quantities as given above.

You must use a cursor for loop for this

Output:

```

100  --Exercise 04
101  BEGIN
102      FOR rec IN (
103          SELECT c.client_id,
104                 i.company,
105                 i.purchase_date
106          FROM clients_obj_tab c,
107                 TABLE(c.investments) i
108      )
109  LOOP
110      IF rec.purchase_date < DATE '2000-01-01' THEN
111
112          UPDATE TABLE(
113              SELECT c.investments
114              FROM clients_obj_tab c
115              WHERE c.client_id = rec.client_id
116          ) i
117          SET i.qty = i.qty + 150
118          WHERE i.company = rec.company
119          AND i.purchase_date = rec.purchase_date;
120
121      ELSIF rec.purchase_date < DATE '2001-01-01' THEN
122
123          UPDATE TABLE(
124              SELECT c.investments
125              FROM clients_obj_tab c
126              WHERE c.client_id = rec.client_id
127          ) i
128          SET i.qty = i.qty + 100
129          WHERE i.company = rec.company
130          AND i.purchase_date = rec.purchase_date;
131
132      ELSIF rec.purchase_date < DATE '2002-01-01' THEN
133
134          UPDATE TABLE(
135              SELECT c.investments
136              FROM clients_obj_tab c
137              WHERE c.client_id = rec.client_id
138          ) i
139          SET i.qty = i.qty + 50
140          WHERE i.company = rec.company
141          AND i.purchase_date = rec.purchase_date;
142
143      END IF;
144  END LOOP;
145
146  DBMS_OUTPUT.PUT_LINE('Bonus Updated Successfully');
147
148  END;
149  /

```

Bonus Updated Successfully



Exercise 05

05. Write a PL/SQL block with an explicit cursor to perform the same set of actions mentioned in exercise 04.

You must use an explicit cursor and a while loop.



PL_SQL_Lab05.sql*

```
150  --Exercise 05
151  DECLARE
152      CURSOR inv_cur IS
153          SELECT c.client_id,
154                 i.purchase_date,
155                 i.company,
156                 i.qty
157          FROM clients_obj_tab c,
158               TABLE(c.investments) i;
159
160      v_client_id  NUMBER;
161      v_date       DATE;
162      v_company    VARCHAR2(20);
163      v_qty        NUMBER;
164
165  BEGIN
166      OPEN inv_cur;
167
168      LOOP
169          FETCH inv_cur INTO v_client_id, v_date, v_company, v_qty;
170          EXIT WHEN inv_cur%NOTFOUND;
171
172          IF v_date < DATE '2000-01-01' THEN
173              v_qty := v_qty + 150;
174          ELSIF v_date < DATE '2001-01-01' THEN
175              v_qty := v_qty + 100;
176          ELSIF v_date < DATE '2002-01-01' THEN
177              v_qty := v_qty + 50;
178          END IF;
179
180          DBMS_OUTPUT.PUT_LINE('Client ID: ' || v_client_id ||
181                               ' Company: ' || v_company ||
182                               ' Updated Qty: ' || v_qty);
183      END LOOP;
184
185      CLOSE inv_cur;
186  END;
187  /
```

Output:

Query result	Script output	DBMS output	Explain Plan	SQL history
 				
Client ID: 1 Company: BHP Updated Qty: 1100 Client ID: 1 Company: BHP Updated Qty: 2000 Client ID: 1 Company: IBM Updated Qty: 700 Client ID: 1 Company: IBM Updated Qty: 1300 Client ID: 1 Company: INFOSYS Updated Qty: 1100 Client ID: 2 Company: INTEL Updated Qty: 500 Client ID: 2 Company: INTEL Updated Qty: 500 Client ID: 2 Company: INTEL Updated Qty: 300 Client ID: 2 Company: FORD Updated Qty: 600 Client ID: 2 Company: GM Updated Qty: 700				